

Multiple Linear Regression Models

STAT 24510-30040

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More than one predictors

— Multiple linear regression models

If there are two predictor variables, the data are in the form

$$(x_{i1}, x_{i2}, y_i), i = 1, \dots, n,$$

and the linear regression model becomes

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i, \quad \varepsilon_i \text{ i.i.d. } \sim N(0, \sigma^2), \quad i = 1, \dots, n$$

Just as in the formulation of the simple linear regression, the linear model is implicitly based on the assumption of conditioning on the predictor variables, so

$$\mathbb{E}(Y_i | X_1 = x_{i1}, X_2 = x_{i2}) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}$$

which is commonly written as

$$\mathbb{E}(Y_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}$$

The *i.i.d* assumption on the ε_i 's means that the model assumes the Y_i 's to be mutually independent, and

$$Y_i \sim N(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}, \sigma^2), \quad i = 1, \dots, n.$$

By the LS method or the ML method, fitting the linear model means finding the best $(\beta_0, \beta_1, \beta_2)$ to fit a plane $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ in $\mathbb{R}^3$ which minimizes the residual sum of squares

$$RSS = \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}))^2$$

among all possible values of $(\beta_0, \beta_1, \beta_2) \in \mathbb{R}^3$, for the observed data $(x_{i1}, x_{i2}, y_i), i = 1, \dots, n$.

There are $p = 3$ coefficient parameters $\beta_0, \beta_1, \beta_2$ estimated by the model.

Under the normality assumption, we can write the likelihood function for the given data.

The MLE's for the β_i 's are the same as the LSE's.

Analogous to the simple linear model, the MLE for σ^2 is

$$\hat{\sigma}_{ML}^2 = \frac{RSS}{n}$$

In fact (general proof later)

$$\frac{RSS}{\sigma^2} \sim \chi_{n-3}^2$$

Thus the MLE $\hat{\sigma}_{ML}^2 = \frac{RSS}{n}$ is biased, and an unbiased estimator for σ^2 is $\hat{\sigma}^2 = \frac{RSS}{n-3}$.

The previous methods we used to derive $\hat{\beta}_i$'s, taking partial derivatives of RSS with respect to each and every β_i to find the MLE or the LSE, still work, but the derivations are getting tedious and complicated as we include more x 's, that is, more explanatory variables.

Fortunately, we have a powerful tool, linear algebra, or rather vector-matrix algebra, to handle the formulation, derivations, and statistical inference of multiple linear regression models.

Using vector-matrix notations in linear algebra, the results for multiple linear regression models actually are cleaner and more concise than the results we painstakingly derived for simple linear regression models.

Vector-Matrix approach for multiple regression models

The simple linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \text{ i.i.d. } \sim N(0, \sigma^2), \quad i = 1, \dots, n.$$

and the linear model with two predictor variables

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i, \quad \varepsilon_i \text{ i.i.d. } \sim N(0, \sigma^2), \quad i = 1, \dots, n$$

are special cases of multiple linear regression model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_r x_{ir} + \varepsilon_i, \quad \varepsilon_i \text{ i.i.d. } \sim N(0, \sigma^2), \quad i = 1, \dots, n$$

The model represents a linear system of n relations,

$$\begin{cases} Y_1 = \beta_0 + \beta_1 x_{11} + \dots + \beta_r x_{1r} + \varepsilon_1 \\ Y_2 = \beta_0 + \beta_1 x_{21} + \dots + \beta_r x_{2r} + \varepsilon_2 \\ \vdots \\ Y_n = \beta_0 + \beta_1 x_{n1} + \dots + \beta_r x_{nr} + \varepsilon_n \end{cases}$$

The i th line corresponds to the i th observation $(x_{i1}, \dots, x_{ir}, y_i), i = 1, \dots, n$.

The linear model can be written more compactly in vector-matrix form

$$Y_{\{n \times 1\}} = X_{\{n \times p\}} \beta_{\{p \times 1\}} + \varepsilon_{\{n \times 1\}}, \quad \varepsilon \sim N_n(0_{\{n \times 1\}}, \sigma^2 I_n)$$

where subscripts indicate matrix dimensions, and I_n is the $n \times n$ identity matrix. The number of coefficient parameters is commonly denoted by $p = r + 1$.

The element-level details of the matrix form are

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}_{[n \times 1]} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1r} \\ 1 & x_{21} & x_{22} & \cdots & x_{2r} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nr} \end{bmatrix}_{[n \times p]} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_r \end{bmatrix}_{[p \times 1]} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}_{[n \times 1]}$$

Recall that the matrix X is called the design matrix of the linear regression model.

Denoting

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}, \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1r} \\ 1 & x_{21} & x_{22} & \cdots & x_{2r} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nr} \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_r \end{bmatrix}, \quad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

The regression model can be written compactly as

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim N_n(0, \sigma^2 I_n)$$

It is equivalent to write the model as

$$Y \sim N_n(X\beta, \sigma^2 I_n)$$

The likelihood function of linear models

The likelihood of the model given observed data $(x_{i1}, \dots, x_{ir}, y_i)$, $i = 1, \dots, n$, is the joint density function of $(Y_1, \dots, Y_n)$, conditioned on $x_{i1}, \dots, x_{ir}, i = 1, \dots, n$.

$$L = L(\beta_0, \dots, \beta_r, \sigma^2) = L(\beta_0, \dots, \beta_r, \sigma^2 \mid (x_{i1}, \dots, x_{ir}, y_i), i = 1, \dots, n) = f_{(Y_1, \dots, Y_n)}(y_1, \dots, y_n)$$

By the independence of the Y_i 's, the joint distribution is the product of the densities of the Y_i 's. Under the normality assumption of the model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_r x_{ir} + \varepsilon_i, \quad \varepsilon_i \text{ i.i.d. } \sim N(0, \sigma^2), \quad i = 1, \dots, n,$$

Thus

$$Y_i \sim N(\beta_0 + \beta_1 x_{i1} + \dots + \beta_r x_{ir}, \sigma^2)$$

The joint density of $(Y_1, \dots, Y_n)$ in the likelihood becomes

$$\begin{aligned} L &= f_{(Y_1, \dots, Y_n)}(y_1, \dots, y_n) = \prod_{i=1}^n f_{Y_i}(y_i) \\ &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2} (y_i - (\beta_0 + \beta_1 x_{i1} + \dots + \beta_r x_{ir}))^2} \\ &= \frac{1}{(2\pi\sigma^2)^{n/2}} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_{i1} + \dots + \beta_r x_{ir}))^2} \end{aligned}$$

The log-likelihood function is

$$\log L = -\frac{n}{2} \log 2\pi - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_{i1} + \dots + \beta_r x_{ir}))^2$$

To use vector-matrix notation, recall that, for vectors

$$v = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}, \quad w = \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} \in \mathbb{R}^n$$

the (Euclidean) inner product of two vectors is defined as

$$\langle v, w \rangle = v_1 w_1 + \dots + v_n w_n = [v_1 \dots v_n] \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} = v^T w.$$

For a vector $w \in \mathbb{R}^n$, its length or the Euclidean norm $\|w\|$ is defined as

$$\|w\|^2 = w^T w = \langle w, w \rangle = \sum_{i=1}^n w_i^2, \quad w = \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} \in \mathbb{R}^n$$

The sum in the likelihood function can be written as

$$\sum_{i=1}^n (Y_i - (\beta_0 + \beta_1 x_{i1} + \dots + \beta_r x_{ir}))^2 = (Y - X\beta)^T (Y - X\beta) = \|Y - X\beta\|^2$$

Thus the likelihood function can be written as

$$L = L(\beta, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{n/2}} e^{-\frac{\|Y - X\beta\|^2}{2\sigma^2}}$$

and the log-likelihood function has the concise form

$$\log L = -\frac{n}{2} \log 2\pi - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \|Y - X\beta\|^2$$

MLE of linear models

The MLE $\hat{\beta}, \hat{\sigma}^2$ are the parameters (a.k.a. arguments) maximizing L thus $\log L$.

$$\{\hat{\beta}, \hat{\sigma}^2\} = \arg \max_{\beta, \sigma^2} \log L = \arg \max_{\beta, \sigma^2} \left(-\frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \|Y - X\beta\|^2 \right)$$

The expression of the $\log L = \log L(\beta, \sigma^2)$ shows that the maximization of $\log L$ with respect to β is independent of the values of σ^2 ,

$$\max_{\beta, \sigma^2} \log L(\beta, \sigma^2) = \max_{\sigma^2} \log L(\hat{\beta}, \sigma^2)$$

and

$$\hat{\beta} = \arg \max_{\beta} \log L = \arg \max_{\beta} \left(-\frac{1}{2\sigma^2} \|Y - X\beta\|^2 \right) = \arg \min_{\beta} \|Y - X\beta\|^2$$

By the way, the last equality shows that the MLE for the coefficient parameter vector β is the same as the LSE, which minimizes $\sum_{i=1}^n (Y_i - (\beta_0 + \beta_1 x_{i1} + \dots + \beta_r x_{ir}))^2 = \|Y - X\beta\|^2$.

Note that for a function $g = g(\beta) = g(\beta_0, \beta_1, \dots, \beta_{p-1})$, the partial derivation with respect to the vector

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_{p-1} \end{bmatrix} \in \mathbb{R}^p$$

is a p vector of partial derivatives with respect to the components of β ,

$$\frac{\partial g}{\partial \beta} = \begin{bmatrix} \frac{\partial g(\beta_0, \beta_1, \dots, \beta_{p-1})}{\partial \beta_0} \\ \vdots \\ \frac{\partial g(\beta_0, \beta_1, \dots, \beta_{p-1})}{\partial \beta_{p-1}} \end{bmatrix}$$

To find the MLE for β , set the p -vector

$$\frac{d}{d\beta} \log L = -\frac{1}{2\sigma^2} \frac{d}{d\beta} \|Y - X\beta\|^2 = 0_p$$

Note that

$$\|Y - X\beta\|^2 = (Y - X\beta)^T (Y - X\beta) = Y^T Y - Y^T X\beta - \beta^T X^T Y + \beta^T X^T X\beta = Y^T Y - 2\beta^T X^T Y + \beta^T X^T X\beta$$

which is a univariate function $\in \mathbb{R}$ of parameters $(\beta_0, \beta_1, \dots, \beta_r)$. Using vector-matrix differentiation (exercise),

$$\frac{d}{d\beta} \|Y - X\beta\|^2 = -2X^T Y + 2X^T X\beta = 0_p$$

Thus we obtain the so called **normal equation** for the MLE $\hat{\beta}$,

$$(X^T X) \hat{\beta} = X^T Y$$

When $X^T X$ is invertible, the LS estimator of the p dimensional parameter vector β can be expressed as

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

The mean vector of $\hat{\beta}$ is

$$\mathbb{E}(\hat{\beta}) = (X^T X)^{-1} X^T E(Y) = \beta$$

The variance-covariance matrix of vector $\hat{\beta}$ is a $p \times p$ matrix,

$$\begin{aligned} Cov(\hat{\beta}) &= Cov((X^T X)^{-1} X^T Y) \\ &= [(X^T X)^{-1} X^T] Cov(Y) [(X^T X)^{-1} X^T]^T = \sigma^2 (X^T X)^{-1} \end{aligned}$$

In the above, we have applied that, for random vector Y and scalar matrices A, B ,

$$Cov(AY, BY) = A Cov(Y) B^T$$

To find the MLE for σ^2 , recall

$$\hat{\sigma}^2 = \max_{\sigma^2} \log L(\hat{\beta}, \sigma^2)$$

and

$$\log L(\hat{\beta}, \sigma^2) = -\frac{n}{2} \log 2\pi - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \|Y - X\hat{\beta}\|^2$$

Taking derivative and set it to zero vector,

$$\frac{d}{d\sigma^2} \log L(\hat{\beta}, \sigma^2) = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \|Y - X\hat{\beta}\|^2 = 0_p$$

We obtain

$$\hat{\sigma}^2 = \frac{1}{n} \|Y - X\hat{\beta}\|^2$$

Recap of multiple linear regression models

The linear regression model is

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim N_n(0, \sigma^2 I_n)$$

An equivalent, useful form for the linear model is

$$Y \sim N_n(X\beta, \sigma^2 I_n)$$

Our interest is on the estimation of β , the goodness of fit of the model, and various inference on the β_i 's.

Results (theorems) on the linear model

- The MLE for the parameter vector β is

$$\hat{\beta} \sim N_p(\beta, \sigma^2 (X^T X)^{-1})$$

- The MLE for the parameter σ^2 is

$$\hat{\sigma}_{ML}^2 = \frac{1}{n} \|Y - X\hat{\beta}\|^2 = \frac{1}{n} RSS$$

- Furthermore,

$$\frac{RSS}{\sigma^2} \sim \chi_{n-p}^2$$

- The parameter estimator and the residual SS are independent:

$$\hat{\beta} \perp\!\!\!\perp RSS$$

- Consequently, t statistics can be constructed to test hypotheses on linear combinations of the β_i 's.

Remarks

We have derived the MLE's in the first two results. The last result is a direct consequence of the definition of the t distribution, based on the third and fourth results. We will prove the the third and fourth results using matrix algebra, especially, the properties of projection matrices.